

Numerical Astrophysics

Current affiliation:

Universidad de Sevilla

US, Sevilla, ESP

Former affiliations:

Pontificia Universidad Católica de Chile

PUC, Santiago, CHL

Universidad Autónoma de Madrid

UAM, Madrid, ESP

University of California

UCSC, Santa Cruz, USA

University of Graz

KF-Uni Graz, AUT

Position: Postdoctoral Researcher

Citizenship: Austrian, EU

CV updated: November 2025

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Experience & Skills

Programming Skills

Python

C & C++ • IDL • bash

SQL • JavaScript • $\LaTeX$

machine learning • HPC

Language Skills

German (native)

English (C1)

Spanish (B2)

Soft Skills

Communication • Outreach

Improvisation Theatre • Dance

Leadership • Mentoring • Teaching

Academic Education

01/2015–12/2019*

UAM, Madrid, ESP

PhD in Theoretical Physics

Instituto de Física Teórica (IFT) & Universidad Autónoma de Madrid (UAM)

10/2005–03/2013

KF-Uni Graz, AUT

Diploma studies (Master) of Astrophysics

University of Graz

Awards & Fellowships

May 2025

Physicists of the Month

Austrian Physical Society

2022–2024

Margarita Salas Fellowship 2021

CA1/RSUE/2021-00720

Ministerio de Universidades

121.100 €

2015–2019

Severo Ochoa Doctoral Fellowship 2014

SEV-2012-0249

Ministerio de Economía y Competitividad

93.150 €

Contracts & Professional Experiences

Jan.–June 2025

Postdoctoral Position in Numerical Astrophysics

University of Sevilla

2020–2021

Postdoctoral Position in Computational Astrophysics

University of California, Santa Cruz (UCSC)

April–Sept. 2013

Software Development

Joanneum Research

Graz, AUT

Freelance contractor at research institute

Further Education & Training (selected)

July–Aug. 2021

Equity-Minded Mentoring

CITL

UCSC, Santa Cruz, USA

Training on mentoring strategies and personal development

Oct. 25–27 2017

Software Carpentry Workshop

IFT-CSIC/UAM

Madrid, ESP

Software Tools and Programming

Sep. 29–Oct. 4 2016

Special Course: SAM Galaxy Formation

IALP

La Plata, ARG

Galaxy formation and evolution modelling incl. software tools

April 25–26 2015

Women in High-Performance Computing

PARC

Dublin, IRL

Initiation in high-performance computing and code parallelisation

July 2011

Software Development

Joanneum Research

Graz, AUT

Internship at research institute

Teaching Experience

March–July 2023

Professor Astronomy (AST0211)

PUC, Santiago, CHL

Sept. – Nov. 2022

Ass. Prof. Astrofísica Experimental (AST0421)

PUC, Santiago, CHL

Advised Theses

Nov. 2025–pres.

Master thesis Astrophysics A. Martínez Morata

VIU, Valencia, ESP

Sept.–Dec. 2022

Graduate Thesis Lic. Astrofísica S. Tancara

PUC, Santiago, CHL

Committees • Mentoring • Leadership

Oct. 2025–pres.

Mentor in SEA programme: Mentoría para Astrónomas

online

2024–pres.

Collaborator Programa Autorreferentes: Mujer y Ciencia

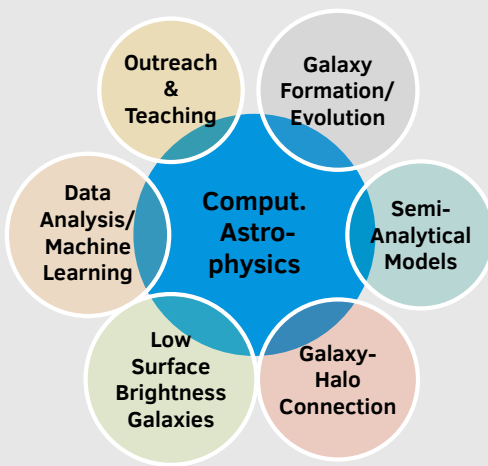
US

July 2023

Committee thesis Lic. Astrofísica V. Miranda

PUC, Santiago, CHL

Research Interests



Project Overview (selected)

FOGLESS+The Machine 2025-pres.
Applying neural networks to cluster galaxy data (in collaboration with Universidad de Complutense, Madrid).
Involvement:
Status:

Hidden Figures in the Sky 2022-pres.
Unveiling the formation and assembly channels as well as the galaxy-halo connection of the in the literature underrepresented *low surface brightness galaxies* using various numerical models of galaxy formation and evolution.
Involvement:
Status:

A SAM Perspective on... 2017-pres.
Paper series focusing on different aspects of galaxy formation and evolution with semi-analytical models.
Involvement:
Status:

The Three Hundred 2017-pres.
Studying 324 central bright cluster galaxies and their evolutionary tracks and galaxy-halo connection through cosmic history.
Involvement:
Status:

- project leader
- main researcher
- support researcher
- data release
- machine learning
- pipelining/coding
- data analysis

Conferences & Workshop Activities (selected)



Dec. 9-14 2024 📍 Puerto Varas, CHL	Deciphering the Cosmic Code for Galaxy Formation	Poster
Oct. 15-19 2024 📍 Sevilla, ESP	X Meeting on Fundamental Cosmology	LOC
July 15-19 2024 📍 Granada, ESP	XVI Reunión Científica de la Sociedad Española de Astronomía	Talk
Nov. 27-Dec. 1 2023 📍 Montevideo, URU	XVII Latin American Regional IAU Meeting	Poster
2019, 2023 📍 OAC, Córdoba, ARG	Friends of Friends Workshop Series	Talk
2019, 2022 & 2023	Galaxy-Halo Connection Collaboration Workshop Series 📍 PUC, Santiago, CHL, 📍 USM, Santiago, CHL, 📍 UDA, Copiapó, CHL	Talk
2019, 2020 & 2023	Mock Workshop Series 📍 OAC, Córdoba, ARG, 📍 UIBK, Innsbruck, AUT, 📍 ULS, La Serena, CHL	LOC, Invited Speaker, Talk

Seminars & Talks (selected)



28/04/2025 📍 DIPIC, Donostia, ESP	Hidden Figures in the Sky Evolution of low-surface brightness galaxies from a hydro-dynamical perspective	DIPIC remote seminars
04/12/2024 📍 ESO, Santiago, CHL	Three years of hiding Insights from studying low-surface-brightness galaxies in the EAGLE simulation within the <i>Hidden-Figures in the Sky</i> -project	TMT Seminar
08/11/2022	A SAM perspective on the challenges of modern galaxy formation modelling	
20/10/2022	Seminario del Departamento de Astronomía	📍 ULS, La Serena, CHL
27/09/2022	Departamento de Astronomía Seminarios	📍 Obs. Cerro Calán, CHL
21/07/2021	Instituto de Astrofísica Colloquia	📍 PUC, Santiago, CHL
05/10/2020	Remote Astro Seminar Series	📍 Uni. of Waterloo, CAN
... 📍 UCSC, Santa Cruz, USA	A SAM perspective on the assembly history and evolution of massive galaxies	UCSC Seminars

Further Scientific Activities



June-Dec. 2026 📍 CAHA, Almería, ESP	Night observations with CAHA iKON 1.23m telescope Project TARSIS	4 nights
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Outreach Activities



Sept. 2024 & 2025	La Noche Europea de l@s Investigad@res	📍 Sevilla, ESP
2024-2025	Programa Autorreferentes: Mujer y Ciencia	📍 Sevilla, ESP
July 2022	Novata por un Día: Visita guiada	📍 PUC, Santiago, CHL
March 2020	Initiative: Astronomía desde el sofá ➡ #AstroEnCasa	online
2019-2021, 2025	Charla con una Astrónoma	online
11/05/2019	IV Jornada de Astronomía en la UAM	📍 Madrid, ESP
Nov. 5-15 2018	XVII Semana de la Ciencia y la Innovación (UAM)	📍 Madrid, ESP
2017-2019	4 videos at the Youtube-channel @IFTMadrid with 124,000 views in total	📍 Madrid, ESP



Publications



In refereed journals:

- [1] [Stoppacher, D.](#) et al. “Hidden figures in the sky: Evolution of low-surface-brightness galaxies from a hydrodynamical perspective”. In: *A&A* 701, A272 (Sept. 2025), A272.
- [2] [Stoppacher, D.](#) et al. “A semi-analytical perspective on massive red galaxies: I. Assembly history, environment, and redshift evolution”. In: *A&A* 693, A219 (Jan. 2025), A219.
- [3] G. Favole et al. “Characterizing the ELG luminosity functions in the nearby Universe”. In: *A&A* 683, A46 (Mar. 2024), A46.
- [4] G. Favole et al. “[OII] emitters in MultiDark-Galaxies and DEEP2”. In: *MNRAS* 497.4 (Oct. 2020), pages 5432–5453.
- [5] [Stoppacher, D.](#) et al. “A semi-analytical perspective on massive galaxies at $z \sim 0.55$ ”. In: *MNRAS* 486 (June 2019), pages 1316–1331.
- [6] Weiguang Cui et al. “The Three Hundred project: a large catalogue of theoretically modelled galaxy clusters for cosmological and astrophysical applications”. In: *MNRAS* 480.3 (Sept. 2018), pages 2898–2915.
- [7] Alexander Knebe et al. “MULTIDARK-GALAXIES: data release and first results”. In: *MNRAS* 474.4 (Mar. 2018), pages 5206–5231.

Further publications:

- [1] A. Castrillo et al. “El observatorio astronómico de la UAM”. In: *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*. July 2020, page 248.
- [2] M. Cortés-Contreras et al. “Astro en casa”. In: *Contributions to the XIV.0 Scientific Meeting (virtual) of the Spanish Astronomical Society*. July 2020, page 253.